

A Hybrid, Density-based Clustering Approach for Spatial Settlement Graphs

Master/Bachelor Thesis/Proposal Presentation

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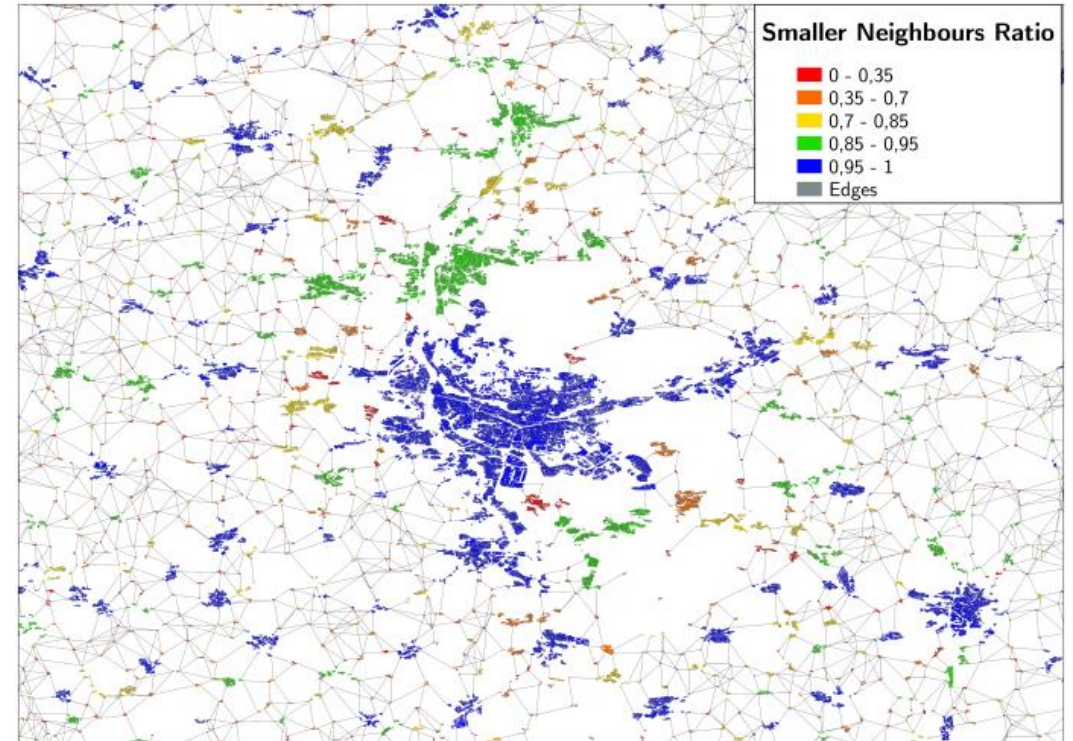
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Motivation

Problem Definition

The delineation of urban settlements from satellite-derived geospatial data remains challenging due to the irregular shapes, spatial fragmentation, and lack of semantic awareness in current clustering algorithms. While graph-based and density-based methods perform well in identifying spatially connected regions, they often fail to consider meaningful attribute similarities, which are crucial for realistic and interpretable clustering in urban contexts.



Settlement chain formed within a single settlement cluster, highlighting how distant built-up areas are incorrectly grouped together [1].



➤ Research questions

- How can alternative clustering algorithms, such as DBSCAN and DBSC, address the issue of settlement chains resulting from the current graph-based clustering approach? Are there other algorithms that could improve compactness and address the limitations of distant settlement connections?
- How can the DBSC algorithm or other identified clustering techniques be effectively integrated into the existing graph-based settlement delineation framework, and which combination of techniques delivers optimal results and improves clustering quality in terms of spatial and semantic coherence?
- What evaluation methods can be employed to objectively assess the quality of the proposed clustering algorithms? Specifically, how can spatial metrics such as compactness and connectivity, computational efficiency, and expert feedback contribute to the assessment of settlement clustering results?



- Several clustering techniques have been explored in geospatial applications:
 - **DBSCAN:** is specifically designed to address the chaining problem seen in earlier clustering methods. It groups points based on density connectivity, which avoids long, thin clusters formed through weak links. However, *DBSCAN does not consider semantic or attribute-based properties* (e.g., population, land use), which limits its effectiveness in geospatial contexts where such attributes are important [2].
 - **GMM:** provides a probabilistic, soft-clustering approach in which each data point is assigned a likelihood of belonging to each cluster, particularly useful for fuzzy boundaries and overlapping regions. However, it assumes that data within each cluster follow a Gaussian distribution, which may not align with the heterogeneous nature of urban settlements [3].
 - **Spectral Clustering:** This method uses the eigenvalues of a graph Laplacian to project data into a low-dimensional space before applying a standard clustering algorithm (e.g., k-means) making it suitable for non-convex and complex cluster shapes [4]. Yet, it requires the entire similarity matrix to be computed, which leads to high computational costs.
 - **Current Graph-Based Method:** constructs a spatial network from geospatial units (e.g., settlement polygons) using proximity or connectivity rules. Model spatial relationships effectively [1]. However, current implementations still produce chaining clusters and lack semantic integration.

Background & Related Work

Motivation



Related Work



Approach



Evaluation



Conclusion

Comparison Clustering Algorithms

Algorithm	Handles noise	Compactness (chaining)	Semantic Attributes	Scalability
DBSCAN	✓	X	X	✓
GMM	X	X	✓	X
Spectral clustering	X	✓	X	X
Graph-based (current)	✓	✓	X	✓
DBSC (proposed)	✓	✓	✓	✓



The proposed method builds on DBSC (Liu et al., 2011):

- Extends the DBSC algorithm to incorporate:
 - **Spatial proximity**
 - **Population similarity**
- Uses a **sweep line algorithm** to compute exact distances between polygon edges [1].
- Clusters are formed when:
 - Distance threshold **AND**
 - Population similarity threshold are both satisfied
- Produces urban clusters that are:
 - **Spatially compact**
 - **Semantically consistent**



Workflow Overview

- **Input Data:** WSF raster converted to polygons
- **Preprocessing:** Filter small patches, enrich with attributes (area, population)
- **Graph Generation:** Build distance matrix between polygon centroids
- **Clustering:** Apply DBSC with distance + similarity constraint
- **Output:** Settlement clusters as shapefiles
- **Evaluation:** Spatial metrics (compactness, connectivity)



To assess clustering quality:

- **Compactness:** Measures the geometric cohesion of clusters, typically via the area-to-perimeter ratio [6]. Higher compactness indicates more physically cohesive urban areas, which better reflect real-world agglomerations.
- **Connectivity:** Assessed through graph metrics such as degree centrality or number of connected components. It reflects how well settlements within a cluster are spatially interlinked, helping detect fragmented or poorly integrated clusters [7].
- **Optional:** Expert validation (if feasible)



- Algorithm Integration and Optimization
- Risks in Data Processing and Resource Limitations with Computing
- Risks in Evaluation Methods
- Expert Surveys as Optional Evaluation Strategy
- Implementation Risks and Future Work
- Mitigation Strategies

Conclusion

Motivation



Related Work



Approach



Evaluation



Conclusion

Problem

Idea

Benefits

Actions

Current clustering methods lack integration of semantic attributes like population, leading to unrealistic settlement clusters and poor spatial coherence.

Adapt the DBSC algorithm to include both geometric proximity and population similarity.

Produces more compact and semantically meaningful urban clusters that better reflect real-world settlement patterns.

Implement the new clustering approach within the existing geospatial pipeline, evaluate with spatial metrics (compactness, connectivity).

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